

Resolution-Clocked Monodromy Counterloops and an Exact Coefficient Bridge

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Abstract

We isolate a deterministic operator-derived counterloop which resolves the simple pole in the Hardy-scaled numerator. For a weighted boundary cycle of length k , edge deletion gives the exact factor $\Pi_{k-1}(\rho_k z^2)$, where $\rho_k \rightarrow \lambda^{-1}$. Its $2(k-1)$ roots have a closed power ledger. Before the first alias at order $2k$, every even moment is $-2\beta_k^n$ and every odd moment is zero. Together with the fixed-order small-noise flat-trace limit this gives an all-order coefficientwise bridge to the deterministic numerator. The object is a graded counterloop, not an identified spectral submultiset of the noisy operator; Gates A–E remain open.

1 Definitions

Put $\lambda = 1.678573510428322\dots$, $r_H = 0.85$, and $\beta = (r_H\sqrt{\lambda})^{-1} = 0.9080523604\dots$. For $k \geq 2$ let ρ_k be the weighted boundary-cycle radius and define

$$\mu_{k,j}^{\pm} = \frac{\sqrt{\rho_k}}{r_H} e^{\pm i j \pi / k}, \quad 1 \leq j \leq k-1.$$

The edge-deflated counterfactor is $C_k(z) = \Pi_{k-1}(\rho_k(z/r_H)^2)$.

2 Exact moment theorem

Theorem 2.1 (monodromy power ledger). *For every $n \geq 1$,*

$$s_{k,n} := \sum_{j,\pm} (\mu_{k,j}^{\pm})^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n), \quad \beta_k = \sqrt{\rho_k}/r_H.$$

Hence $s_{k,n} = -2\beta_k^n$ for even $n < 2k$ and $s_{k,n} = 0$ for odd $n < 2k$.

Proof. Pair conjugates and sum the cosines. The sum over $j = 1, \dots, k-1$ is -1 unless $k \mid n/2$, in which case it is $k-1$; the odd case cancels under $j \leftrightarrow k-j$. This gives the displayed indicator formula. $\square$

Proposition 2.2 (coefficientwise bridge). *Let $c_{\sigma,n}$ be the Hardy-scaled flat trace after the two peripheral branches are removed and suppose $c_{\sigma,n} \rightarrow r_H^{-n} c_n$ for every fixed n . If $k = k(\sigma) \rightarrow \infty$ and $\rho_k \rightarrow \lambda^{-1}$, then for each fixed $n \geq 2$,*

$$c_{\sigma,n} - s_{k(\sigma),n} \longrightarrow r_H^{-n} c_n + 2\mathbf{1}_{2|n} (r_H\sqrt{\lambda})^{-n} = a_n.$$

Proof. For sufficiently large k , $n < 2k$, so the theorem applies. Then take the two stated limits and use the deterministic numerator dictionary. $\square$

3 Local-uniform model quotient

Write the Hardy-scaled deterministic bulk factor as $\widehat{D}_{0,H}(z) = G_H(z)/(1 - (\beta z)^2)$, where G_H is the holomorphic numerator fixed by the deterministic coefficient dictionary. If $\rho_k = \lambda^{-1}$ exactly, then

$$C_k(z) = \frac{1 - (\beta z)^{2k}}{1 - (\beta z)^2}.$$

Thus $\widehat{D}_{0,H}$ divided by C_k equals $G_H(z)/(1 - (\beta z)^{2k})$. For $R < \beta^{-1}$ its relative error from G_H is at most $(\beta R)^{2k}/(1 - (\beta R)^{2k})$. The same conclusion holds with $\rho_k \rightarrow \lambda^{-1}$ on every strict subdisk after a separate radial perturbation estimate.

Remark 3.1. This is an exact graded/superloop bridge. RH-242 permits the finite atomic counterloop for any multiset, but ordinary determinant quotient language is valid only when the multiset is an actual spectral submultiset. That noisy identification is not proved here.

4 Boundary

The counterloop branch supplies a strict anchored head and a model tail interface. It does not supply a moving-cloud envelope, a common noisy contour, a Hilbert–Polya operator, zeta-divisor equality, or RH.